SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CS A5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks: 100 Annexure A

CONCEPT MAPPING

Code of Conduct:

I T Sanhith (192311228) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

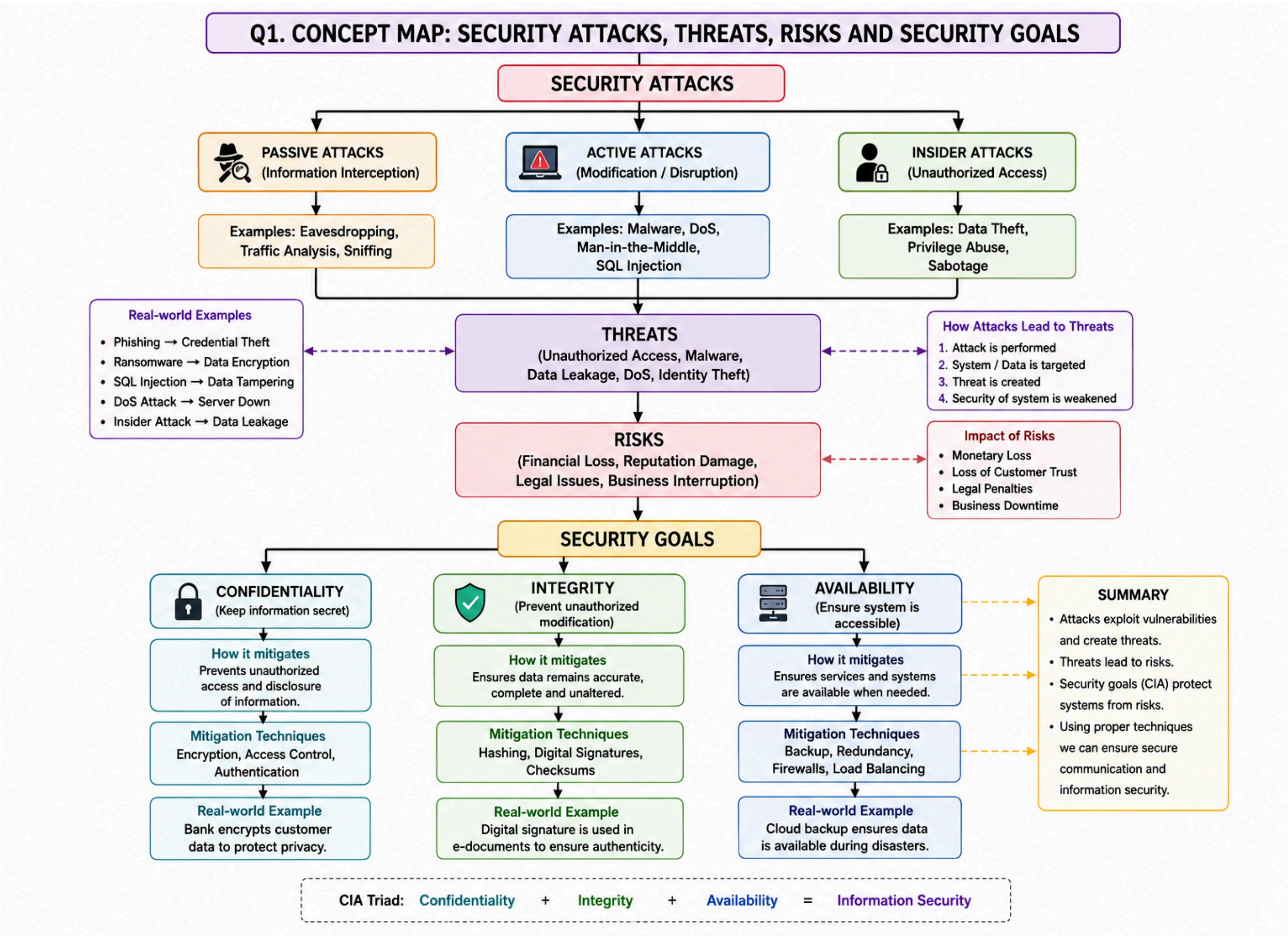
Signature of the student:

Q. No	*Understanding of Concepts (25)*	*Experimental Design (30)*	*Use of Tools (20)*	*Analysis of Results (15)*	*Documentation (10)*	*Total Score (out of 100)*
1	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
2	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
3	*/ 5*	*/6*	*/ 4*	*/ 3*	*/ 2*	
4	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
5	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
OVERALL RUBRICS VALUE						
	/ 5	*/ 5*	*/ 4*	*/ 4*	*/ 2*	
	/ 25	***/ 30***	***/ 20***	***/ 15***	***/ 10***	***/ 100***

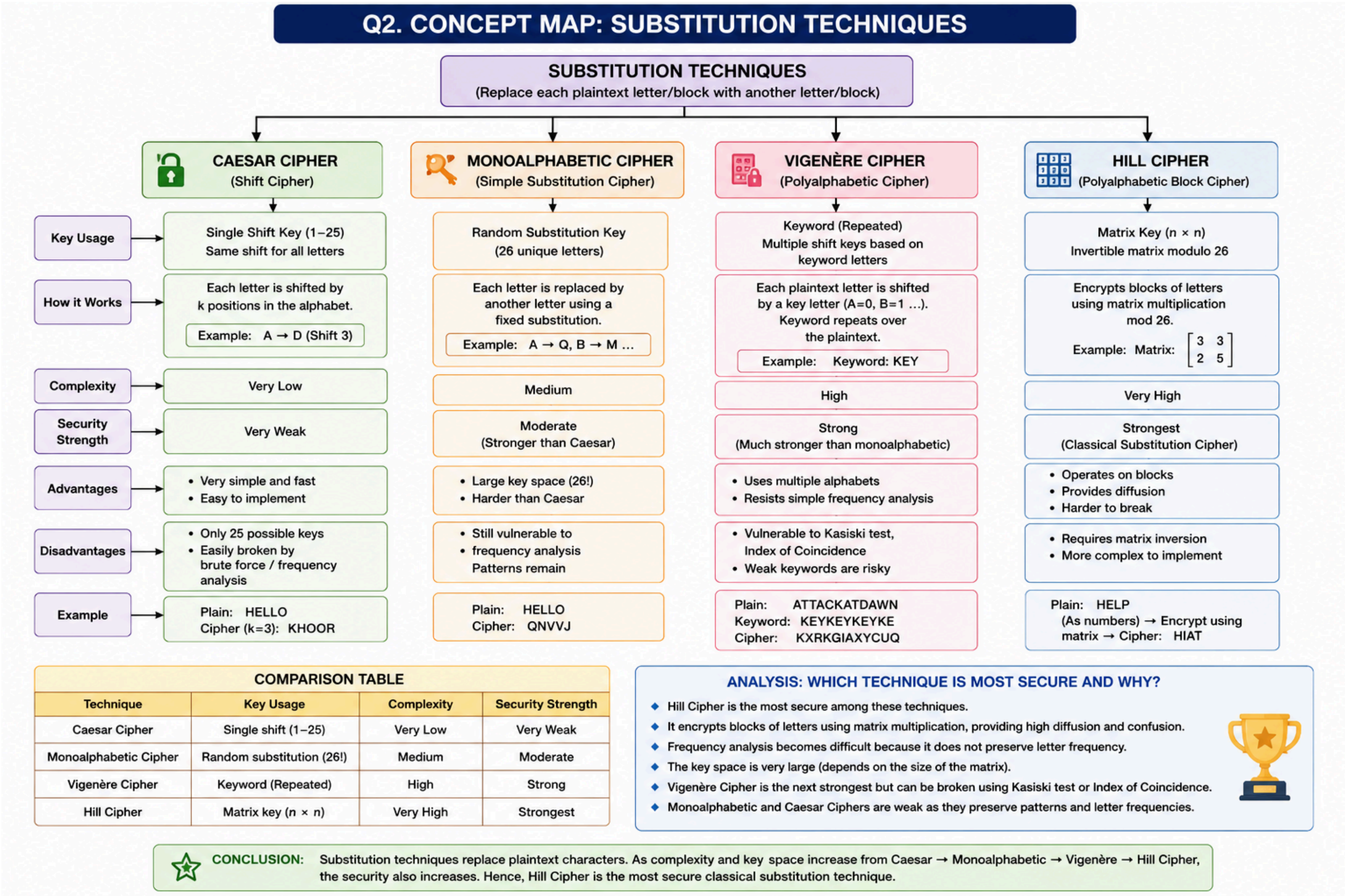
Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link



2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.



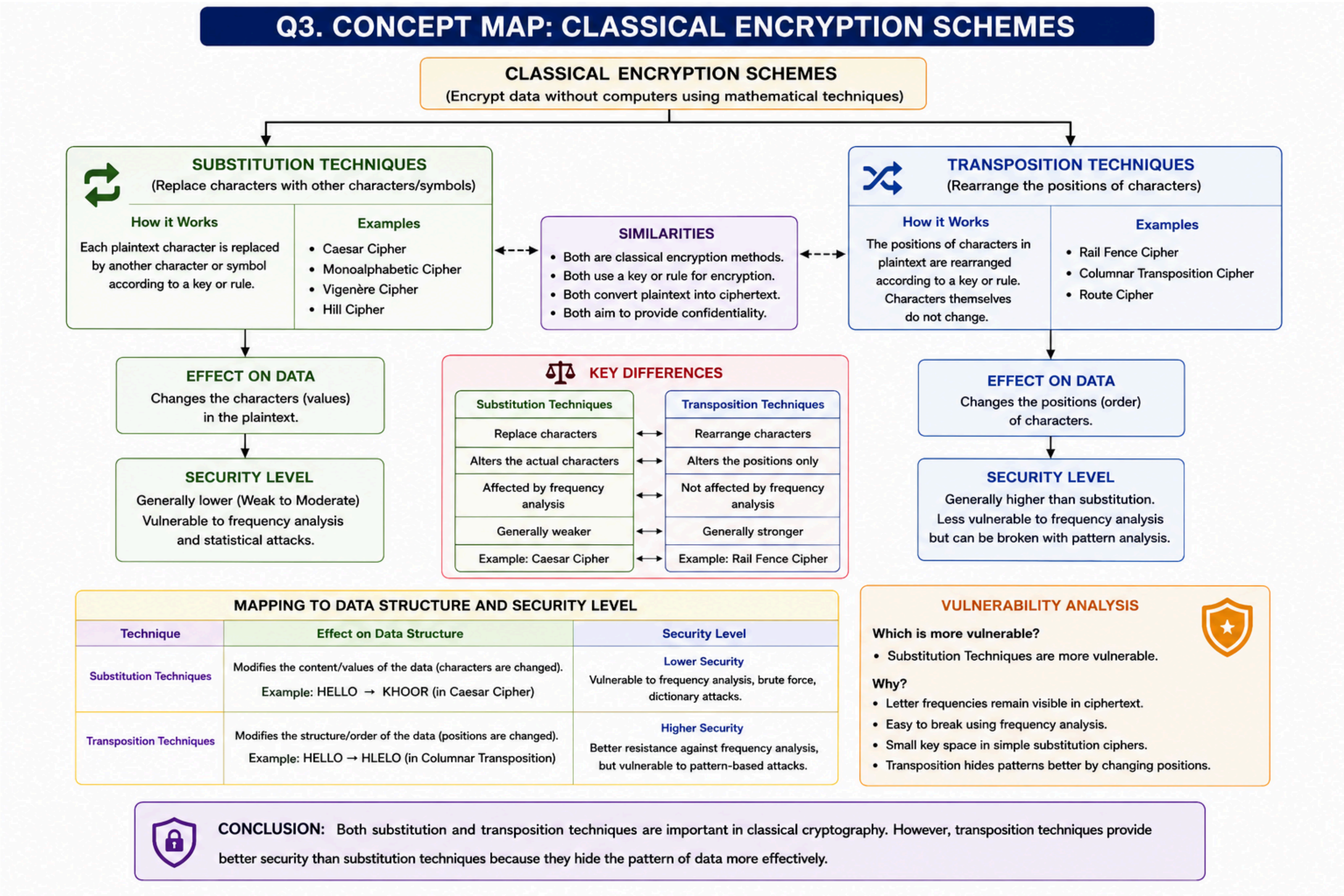
Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	Single shift (1–25)	Very Low	Very Weak
Monoalphabetic Cipher	Random substitution (26!)	Medium	Moderate
Vigenère Cipher	Keyword (Repeated)	High	Strong
Hill Cipher	Matrix key (n × n)	Very High	Strongest

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why

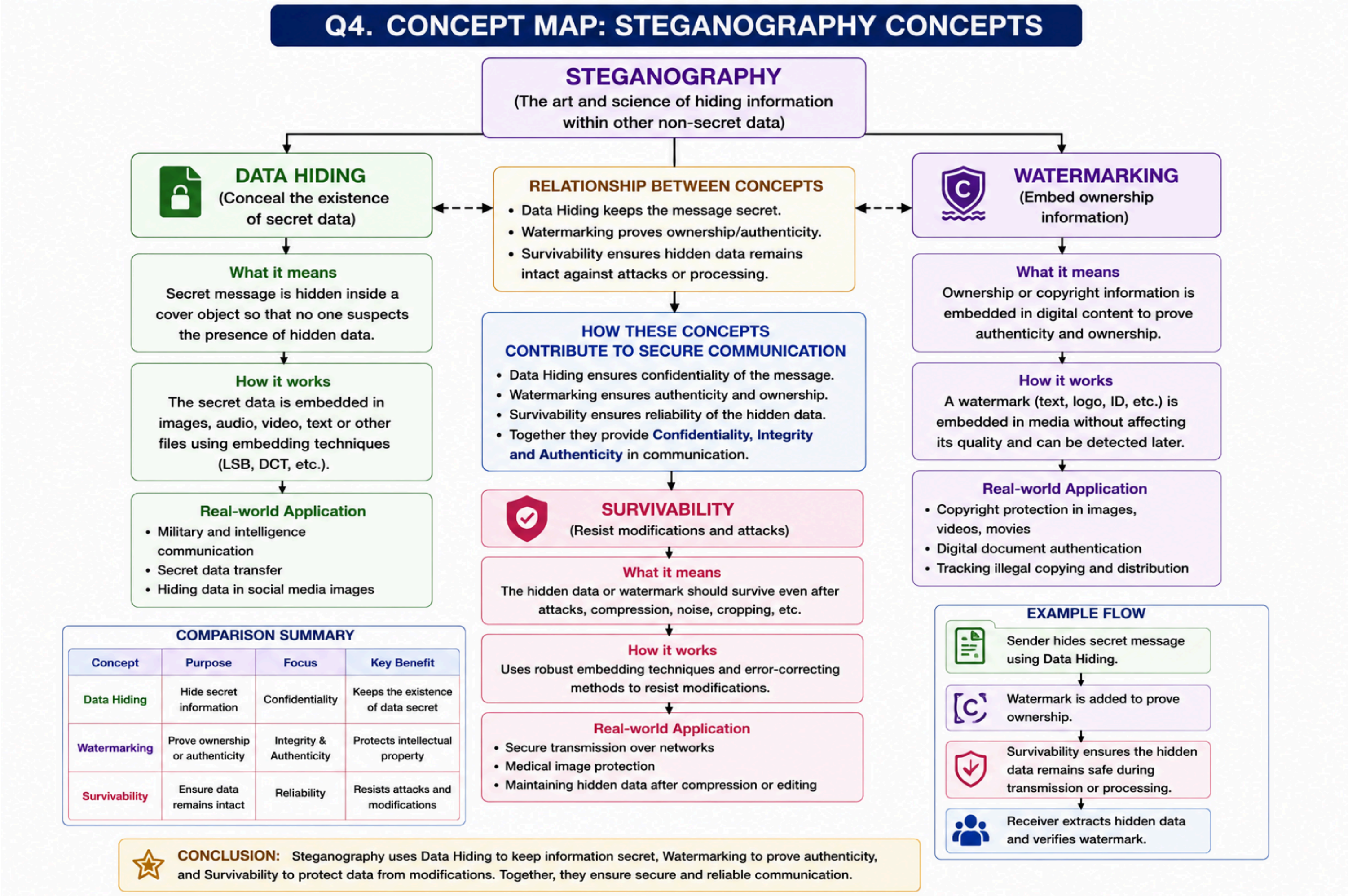


4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

a. Show relationships between these concepts

b. Analyze how each contributes to secure communication

c. Include one real-world application for each



5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks: a. Show how each concept is mathematically connected

b. Map their role in encryption/decryption

c. Analyze which concept is most critical for public key cryptography and justify